Chain-NPM: Technical Appendix

Supplementary material for the submitted paper

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ABSTRACT

This companion holds the material that the page-limited submission cannot carry: the witness table for the identifiability barrier, the Phase-2 walk-through, the numeric tables behind the controlled-benchmark figure, the deep TPC-H and ACS rails, the planted-effect and natural-mode tables, the three real-benchmark rails figure, and the budget-allocation, ancestor-scale and cost–effect ablations. The submission cites each item where it is relevant; nothing here adds a new claim. Section, figure, table and proposition numbers written as "of the submission" refer to the paper and are resolved at build time.

1 SENSITIVITY CONSTANTS AND TOTAL-VARIATION STATEMENTS

Proof of Lemma 5.1. Fix neighboring databases D, D' differing in a single individual tuple $i^\star$ of group $g^\star$ (size $s = \min(|g^\star|, \tau) \geq 1$). Every released marginal is a sum over groups of a per-group normalized contribution, and only the $g^\star$ term differs. *(i) Intra-table 1-way.* The affected count changes by 1, and the public-free normalization denominator by 1; the ℓ_2 change of the vector is ≤ 1. *(ii) Adjacent- and cross-hop group-normalized marginals.* The per-group term is $c_g = \frac{1}{s_g}\sum_{i\in g} x_i$ with x_i a one-hot attribute vector. Adding $i^\star$ (with $s < \tau$) changes $c_{g^\star}$ by $\|x - c\|_1/(s+1) \leq 2/(s+1) \leq 1$; with $s = \tau$ the clipped count changes by $\leq 1/\tau \leq 1$; removing $i^\star$ changes it by $\|x - c\|_1/s \leq 2/s$, which equals 1 when $s = 1$ (the group vanishes) and is ≤ 1 otherwise. Hence the ℓ_2 sensitivity is ≤ 1. *(iii) Within-group pair marginals.* The per-group pair term changes only through pairs involving $i^\star$: at most $2(s-1)/s^2 \leq 1$ in mass, plus a renormalization term ≤ 1 when s crosses the truncation; the worst case ($s = 2$ removal) gives ℓ_2 change ≤ 2. Since only one group term moves in each case, the stated bounds are the ℓ_2 sensitivities. □

Lemma 1.1 (Tight ℓ_2 sensitivities). *Under the one-individual adjacency of Lemma 5.1, let s be the truncated size of the affected group. Then: (i) intra-table 1-way marginals have $\Delta_2 = 1$; (ii) adjacent- and cross-hop group-normalized marginals have $\Delta_2 = 1$, attained by removal of a single-individual group, while any addition satisfies $\Delta_2 \leq 2/(s+1) < 1$; (iii) within-group pair marginals have $\Delta_2 = 2$ in the statement of Lemma 5.1, with the maximizing case a removal at $s = 2$; for $s \geq 3$ the per-group change is at most $2(s-1)/s^2 \leq 3/4$ plus a $O(1/s^2)$ renormalization term.*

Proof. The three cases are precisely the per-group computations of Lemma 5.1; the values above record the supremum over s and over add/remove together with explicit witnesses (single-individual group removal for (i)–(ii), two-individual group removal for (iii)), hence the bounds are tight. □

Corollary 1.2 (Total variation of released marginals). *Let $M \in \mathbb{R}^d$ be any marginal of Lemma 5.1 with ℓ_2 sensitivity Δ_2, released as $\hat{M} = M + \mathcal{N}(0, \sigma^2 I)$. With probability $\geq 1-\beta$, $\|\hat{M} - M\|_\infty \leq \sigma\sqrt{2\ln(2d/\beta)}$; if M is normalized by a group count of magnitude $n_1 m_{\min}$, then $\mathrm{TV}(\hat{M}/\|\hat{M}\|_1,\ M/\|M\|_1) \leq \frac{d\,\sigma}{n_1 m_{\min}}\sqrt{2\ln(2d/\beta)}$.*

Proof. A union bound over the d Gaussian coordinates gives the ℓ_∞ statement. Total variation is at most half the ℓ_1 difference of the normalized vectors, hence at most $\frac{d}{2}\|\cdot\|_\infty$; dividing by a mass of magnitude $n_1 m_{\min}$ converts absolute to relative error, absorbing the normalization perturbation into the same order. □

Lemma 1.3 (TV decay along the chain of posteriors). *Consider a k-hop chain with true channels $P_{Z_i|Z_{i-1}}$ and synthesized channels $\hat{P}_{Z_i|Z_{i-1}}$ satisfying $\max_{z_{i-1}} \mathrm{TV}(\hat{P}_{Z_i|z_{i-1}}, P_{Z_i|z_{i-1}}) \leq \eta_i$. Then the synthesized endpoint joint satisfies $\mathrm{TV}(\hat{P}_{Z_0,Z_k}, P_{Z_0,Z_k}) \leq 1 - \prod_{i=1}^k(1-\eta_i) \leq \sum_{i=1}^k \eta_i$, and the recoverable endpoint correlation obeys $|\Delta_{\mathrm{syn}}| \leq \prod_i(1-\eta_i)\,\Delta_{\mathrm{real}} + 2\sum_i \eta_i$. Each η_i is lower-bounded by the minimax TV risk of estimating the hop's channel from its budget slice under the Gaussian mechanism, hence is bounded away from 0 at any fixed total budget.*

Proof. Write the joint as a Markov chain and telescope: replacing the i-th channel changes the joint distribution by at most η_i times the mass that reaches hop i, which is at most the product of the survival factors $(1-\eta_j)$ of the preceding hops plus the channel error itself; induction yields the product form, and the data-processing inequality for TV gives the additive relaxation. The correlation bound follows by writing Δ as a linear functional of the endpoint joint and bounding by TV. The minimax claim is the standard Gaussian-mean lower bound (Le Cam's method applied to the channel entries at the hop's γ_i slice). □

2 PLANTING PROTOCOL, V-SHAPED SENSITIVITY, AND STATISTICAL CONVENTIONS

Planting protocol. To instantiate the barrier of Proposition 5.5 on real schemas, we plant cross-hop structure while leaving every one-hop object untouched. On a chain $R_0 \leftarrow R_1 \leftarrow R_2$ we pick two

balanced root anchors X_1, X_2 and set the latent $Z = X_1 \oplus X_2$ (pure interaction, no main effects) under *plant_xor*, or $Z = X_1$ under *plant_product*. The leaf response A is rewritten with probability p by the rule ($A \sim \mathrm{Bern}(r_Z)$ or $A = U \oplus Z$) and kept real otherwise; then, within each value class of a middle attribute, count-preserving swaps restore the (R_1-attr $\times$ A) contingency exactly. Hence the FK cardinalities and every one-hop marginal the per-FK family consumes are unchanged by construction, while the cross-hop functional moves by a calibrated amount; p is chosen by bisection over seeds so that the planted effect hits $\Delta \in \{0.10, 0.20, 0.316, 0.40\}$ under *plant_product*, or the XOR target value 0.8 under *plant_xor*. Repeating the protocol with the rule family $\{\oplus, \mathrm{mod}k, 1[\cdot \geq t], \vee\}$ forms the non-linearity scan.

V-shaped sensitivity. For a star $R_1 \leftarrow C \rightarrow R_2$, the star workload adds the cross-parent marginal $R_1 \times R_2$ and the triple $R_1 \times R_2 \times C$, each child row weighted by the inverse of its R_1-group size (truncated at τ). For a neighbor obtained by adding or removing one R_1 tuple together with its C rows, each star query's ℓ_2 sensitivity is at most 1 by the per-group argument of Lemma 5.1; the R_2-side neighbor follows by the same argument with roles swapped.

Statistical conventions. Ten seeds per configuration with common random numbers (data, query-workload, and planting seeds shared across methods); medians with bootstrap 95% confidence intervals; per-seed paired Wilcoxon signed-rank tests with Holm correction over the family "Chain-NPM vs. each baseline" per dataset, ε, and metric. Seeds, ε grids, and metric definitions are fixed before baseline tuning and released with the artifact.

3 THE DEEP-CHAIN NORMALIZATION STUDY

Measuring a cross-hop marginal is not sufficient on its own; two normalization choices matter, and both show up as failures of effect retention. (i) The cross-hop marginal must be normalized *once*, by the FK group it is traversed through. Normalizing a second time by the ancestor group collapses the released cross-hop mass from 54,369 to 1,000 while the noise is unchanged, leaving the measured object at signal-to-noise ratio $O(1)$—and the resulting retention error returns to 1.145, i.e. *worse* than not measuring the marginal at all (1.027), against 0.380 under the rule the paper adopts. (ii) The lift denominator is itself a noised one-way marginal of a small table, so an unguarded cross-hop lift manufactures a spurious contrast of +0.15–+0.20 at $\varepsilon \leq 0.4$ where the true contrast is ≈ -0.004; the existing SNR shrinkage cuts that artefact by a factor of 1.9–4.9 at ε=0.2–0.8. It is not a cure everywhere: at ε=0.1 the guard barely moves the artefact (+0.197 $\rightarrow$ +0.177), because the noised root marginal sits far above the noise scale, and the natural-mode two-hop error there also worsens (3.88 $\rightarrow$ 4.34, both measured without the quota calibration), so at ε=0.1 the cross-hop object costs more than it returns. The row-level quota calibration of 6.1 addresses exactly this cell: the natural two-hop error drops from 4.34 to 0.28 without changing a released marginal. With both rules the planted effect on the deep chain is recovered (1.027 $\rightarrow$ 0.380), and the guard is inert on the other three datasets, whose V-shaped path already measured its cross-parent object. Two further costs of the rule are worth stating plainly: the twelve extra cross-hop marginals raise the released-marginal count from 31 to 43 and, under an equal budget split, the noise on every query by 17%; and where the guard does not bite the extra marginals dilute accuracy, as the deep TPC-H rail below shows.

Table 1: A minimal instantiation of the identifiability barrier (5.5). Two databases $\mathcal{D}^+$ and $\mathcal{D}^-$ over the same three-level chain (2 regions $\times$2 households $\times$2 individuals per household, uniform group sizes) share *every* per-FK view—all table marginals are uniform, $Z \perp U$ makes the $R_0 \rightarrow R_1$ view coincide, and the within-group composition of the $R_1 \rightarrow R_2$ view is identical because the sign of A is symmetric in U (the Z=0 households swap roles between the two databases). Yet the two-hop join functional $\varphi = \Pr[A = U \oplus Z]$ equals 1 on $\mathcal{D}^+$ and 0 on $\mathcal{D}^-$. Any synthesizer whose input is the multiset of per-FK views receives identical input in the two cases and therefore errs by at least 1/2 on φ for one of them, at any budget.

U (R_0)	Z (R_1)	A in $\mathcal{D}^+$	A in $\mathcal{D}^-$
0	0	0	1
0	0	0	1
0	1	1	0
0	1	1	0
1	0	1	0
1	0	1	0
1	1	0	1
1	1	0	1

4 THE IDENTIFIABILITY BARRIER ON A WITNESS CONSTRUCTION

Proposition 5.5 of the submission is proved by a two-database witness that a reader can verify row by row; the submission states the barrier and defers the table to this companion.

5 PHASE-2 WALK-THROUGH

The submission describes the EM soft assignment and the ancestral sampler in prose; the figure below traces one refinement-and-sampling step on the running example's micro-instance.

6 NUMERIC TABLES FOR THE CONTROLLED MULTI-HOP BENCHMARK

The submission reports the controlled benchmark of Section 6.3 of the submission as a figure (Figure 6 of the submission) and quotes the per-cell numbers in its prose. The two tables below are the same runs and medians in tabular form.

7 TPC-H RAILS (STANDARD AND DEEP)

Standard rail: what TPC-H does and does not separate.

We use the TPC-H benchmark (SF=1) with the correlation-planting transformations of [2] (adopted by PrivPetal [3]): (i) lineitem prices/quantities are scaled by part-type and part-brand weights; (ii) each year's orders are concentrated on a random

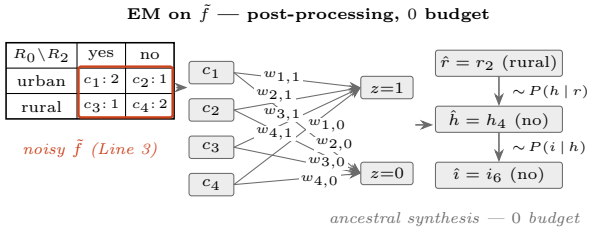


Figure 1: Walk-through of Phase 2 on the running example's micro-instance: the noisy cross-hop marginal $\tilde{f}$ (left, Line 3) is fed to the EM soft assignment of each cell c to the latent household type $z \in \{1, 0\}$ with weights $w_{c,z}$ (middle), and the fitted conditionals then drive ancestral synthesis $\hat{r} \to \hat{h} \to \hat{\imath}$ (right). Both stages operate solely on the released noisy statistics—pure post-processing—and therefore consume zero additional privacy budget (Algorithms 2–3).

Table 2: Two-hop Δ-preservation ratio $\Delta_{\text{syn}}/\Delta_{\text{real}}$. Per-FK family stays within the ±0.1 noise band of the no-cross-table level at every ε; the joint reference is consistently positive. The PrivLava row is its released code; a faithful implementation of its Algorithm 3 stays within 0.06 of it in these cells.

Method	ε=0.8	ε=3.2	ε=8.0
PrivMRF (lower bound)	0.00	−0.02	−0.09
PrivLava	0.02	0.01	0.03
PrivMRF (denorm) (joint ref.)	0.21	0.29	0.21

Table 3: Query relative error (large-count queries, ε=3.2), reported per hop class (1-hop R⋈H, 1-hop H⋈I, 2-hop R⋈H⋈I), for each method, with the two Chain-NPM variants grouped below the baselines. PrivLava is its released code; a faithful implementation of its Algorithm 3 stays within 0.04 of it here.

Method	1-hop RE (R⋈H)	1-hop RE (H⋈I)	2-hop RE (R⋈H⋈I)
PrivMRF	0.224	0.151	0.209
PrivMRF (denorm) (joint ref.)	0.020	0.321	0.420
PrivLava	0.246	0.165	0.351
Chain-NPM-direct	0.058	0.084	0.037
Chain-NPM-mixture	0.049	0.030	0.114

15% subset of customers, coupling *customer nation* to *order year*; (iii) orders are dropped unless their size matches the order year, coupling *order size* to *order year*. Private relations: Orders and Lineitem; public: Customer, Nation, Part. This yields 536K orders and 2.1M lineitems. Ground truth for evaluation is the planted database converted back to the original schema (identical to PrivPetal's own postprocessing).

Chain-NPM runs on this schema with the nation attribute joined into Orders (public joins are free), Q=42 joint chain marginals in one budget pass, $\delta = 1/n_{\text{leaf}}$. PrivPetal is run with its official code (tuple_num=2, process_num=4). Metrics: median relative error of multi-hop conjunctive count queries (2-hop Nation⋈Orders⋈Lineitem and 1-hop Orders⋈Lineitem), and preservation ratios of the planted associations.
Additional planted-association preservation (ratios, ε=0.2 / ε=3.2): order-size↔year 1.02 / 0.98 (PrivPetal), 1.16 / 1.16 (Chain-NPM); mean lineitem quantity 1.01 / 1.01 (PrivPetal), 0.98 / 0.96 (Chain-NPM). PrivPetal shows noticeable run-to-run variance at low budget (a repeat ε=0.2 run gave 2-hop RE 0.072 and 1-hop RE 0.019); we report the clean rerun and note the stochasticity.

Discussion: what TPC-H does and does not separate. On the standard TPC-H setup, PrivPetal performs well—and so does Chain-NPM. This is expected and informative: with only *two* private relations (Orders, Lineitem) and all upstream tables public, every planted correlation lives within a single FK pair after the public join (the nation↔year association is carried by Orders' foreign key to the public Customer table), so per-FK models *can* express it. The two methods are indeed comparable here (per-cell numbers in the technical appendix): Chain-NPM has a slight edge on two-hop query error at low budget (0.044 vs 0.083), PrivPetal a slight edge at high budget (0.015 vs 0.031), and both preserve all planted associations. The structural gap of Section 3.1 instead requires *deep private chains*—as in the controlled three-table benchmark of Section 6.3, where the per-FK family's Δ-recovery stays at ≈ 0 regardless of budget. The next paragraph tests this on the real schema.

Deep-chain variant: Customer private. A deep variant with a *three*-level private chain Customer → Orders → Lineitem—the real-schema instance of the structure Section 3.1 identifies—is reported in full in the technical appendix. Chain-NPM answers 3-hop queries at 5.4% median relative error at ε=3.2 against PrivPetal's 4.0%, and pays for the cross-hop object at the low budget (33.6% against 12.6%): its twelve extra marginals dilute the same single budget. Where it differs from the latent-chain baseline is cost and coverage—≈ 22–24 seconds of synthesis on one CPU against 1.7–2.3 hours per budget on an A800 GPU, one budget pass over 46 marginals rather than a three-way split, and all 150K customers synthesized, including the 50K without orders, where PrivPetal emits only the 99,847 with orders.

Efficiency. Chain-NPM's entire pipeline (one budget pass over 42 joint marginals plus post-processing synthesis of 536K orders and 2.1M lineitems) completes in ≈ 2.2 minutes on a single CPU. PrivPetal's official implementation takes 1.8–2.1 hours per budget on two A800 GPUs for the same schema—a ≈ 50× wall-clock gap, which the hardware asymmetry makes conservative for Chain-NPM while our exclusion of the evaluation from the timed window shades it the other way—mirroring the single-pass-vs-iterative contrast on the controlled benchmark and following directly from EM being post-processing rather than a budget-consuming loop. The deep-chain variant shows the same pattern: 1.7–2.3 hours per budget on one A800 for PrivPetal versus ≈ 22–24 seconds for Chain-NPM.

Standard rail (Orders and Lineitem private).

Table 4: TPC-H (planted, SF=1). Δ-ratio = preserved/true planted association (1.0 = perfect); RE = median relative error of multi-hop count queries (2-hop Nation×Orders× Lineitem).

	ε=0.2		ε=3.2	
Method	2-hop RE	nat↔yr	2-hop RE	nat↔yr
PrivPetal (official)	0.083	1.05	0.015	0.99
Chain-NPM (ours)	0.044	1.04	0.031	1.01

Table 5: Deep TPC-H (Customer private, SF=1). 3-hop RE = median relative error of Nation×Segment×Year×Qbin join counts; seg↔year is the three-table composition signal (preserved/true, 1.0 = perfect).

	ε=0.2		ε=3.2	
Method	3-hop RE	seg↔yr	3-hop RE	seg↔yr
PrivPetal (official)	0.126	1.12	0.040	0.98
Chain-NPM (ours)	0.336	1.72	0.054	1.73

Deep rail (Customer private). We build a deep TPC-H variant with a *three*-level private chain Customer → Orders → Lineitem: Customer carries nation, market segment, and account-balance-bin attributes; the existing plantings (nation↔year across Customer→Orders, size↔year and part-ratio-scaled quantities across Orders→Lineitem) are retained, and a new within-Customer segment↔nation association is planted, so the composition segment→nation→year→quantity spans all three private tables. Evaluation adds 3-hop conjunctive counts (Nation ×Segment×Year×Quantity-bin) over the 3-table join. Chain-NPM extends to a three-level chain (Q=46 marginals, one budget pass, intermediate rows sampled once and conditioning both hops, by the construction of Section 4.4 of the submission); PrivPetal runs its official multi-stage pipeline (budget split 0.5/0.25/0.25 over its Customer–Orders, Orders–Lineitem, and Partsupp–Lineitem stages).

Both methods handle the deep chain. PrivPetal's multi-stage latent pipeline answers 3-hop queries at 4.0% median relative error at ε=3.2 (0.126 at ε=0.2) and preserves all planted associations (ratios 0.98–1.12). Chain-NPM attains 0.054 at ε=3.2 but pays for the cross-hop object at the low budget (0.336): its twelve extra marginals dilute the same single budget (6.2), so at ε=0.2 the latent-chain baseline is ahead on query error while Chain-NPM remains ahead on cost and coverage. This contrast is honest and informative: PrivPetal is itself a chain-aware model—its child stage is conditioned on the synthetic parent rows, so composed correlations propagate—and the structural collapse of Section 3.1 of the submission is the failure mode of the PrivLava-style per-FK family (Table 2), not of PrivPetal-style latent chains. Where Chain-NPM differs is cost and coverage: the deep rail's synthesis completes in ≈ 22–24 seconds on a single CPU versus 1.7–2.3 hours per budget for PrivPetal's own pipeline on an A800 GPU (≈ 300× on the bases stated in 8); it consumes one budget pass over 46 marginals rather than a three-way split; and it synthesizes all 150K customers, including the 50K without orders, whereas PrivPetal emits only the 99,847 customers with orders (customer-only segment counts then incur RE 0.33). Chain-NPM's segment↔year ratio slightly exceeds 1 because the absolute composition signal is small (total-variation 0.33 across 35 cells) and both noise and lift-composition inflate small values; the 3-hop query errors are the primary evidence. Deep chains are intrinsically harder at low budget: at ε=0.2 the single pass must spread $\sigma \approx 194$ Gaussian noise over 46 marginals whose effective mass is set by the 100K customer level rather than the 536K order level. Two of the composition ratios thereby leave the band we would like—the mean-lineitem-quantity ratio reaches 1.46 at ε=0.2, against 1.00 at ε=3.2—and the synthesized order and lineitem counts shrink by 6–8%; the correlation structure itself is retained. This is the same correlations-cheap, counts-expensive pattern as on the controlled benchmark, seen from the other side.

8 ACS PUMS CALIFORNIA

The rail.

To verify that Chain-NPM works beyond engineered benchmarks, we run it on the ACS PUMS 2022 California microdata: 151K occupied households and 383K persons (vacant units excluded), with 7 discretized attributes on each table (household type/tenure/income/structure, person age/sex/education/employment/relationship/hours). This is a natural two-level chain with heavy-tailed group sizes (max 20) and strong size–attribute coupling—the stress case for the size model. Chain-NPM uses Q=74 marginals (7 one-way per table, 5 parent pairs, 49 parent×child cells, 6 size histograms) with the size-itself attribute excluded from size conditioning; PrivPetal runs its official Census pipeline.

Chain-NPM answers 2-hop conjunctive counts at ≈ 20% median relative error (large counts) and person-level marginals at around 10% at both budgets (the technical appendix tabulates the rail), with household-size totals within 7% of truth; 0-hop household marginals stay below 2%. This trails the latent baseline: PrivPetal reaches 5.8–9.0% 2-hop and 1.5–2.0% 1-hop relative error with nearly exact counts (−0.7%), at the price of hours of GPU time per budget. The score is informative. The controlled benchmark of Section 6.3 isolates the *structural* per-FK failure; ACS has no planted correlations to lose, so both methods are learning the same 7×7-attribute joint, and PrivPetal's per-stage latent models simply fit a two-level join more efficiently than our single-pass pairwise marginals—a budget-allocation effect, not a structural one. Two-hop joins over a 7×7 attribute product with only 151K parent rows are noise-limited at these budgets, and the contrast confirms the mechanism's limitations on skewed, heavy-tailed microdata while an exactly-calibrated size model and a single-pass pipeline retain their cost and simplicity advantages. Chain-NPM uses Q=74 marginals (7 one-way per table, 5 parent pairs, 49 parent×child cells, 6 size histograms) with the size-itself attribute excluded from size conditioning; PrivPetal runs its official Census pipeline.

Table 6: ACS PUMS California 2022. RE (6.1) on random conjunctive counts with true answer ≥ 50 (2-hop: household× person over the join; 1-hop: person-only).

	ε=0.8		ε=3.2	
Method	2-hop RE	1-hop RE	2-hop RE	1-hop RE
PrivPetal (official)	0.090	0.020	0.058	0.015
Chain-NPM (ours)	0.196	0.105	0.198	0.102

9 ABLATIONS AND EFFICIENCY

Budget allocation across marginals. We ablate the equal γ^2-split by re-weighting the seven cross-hop marginals (four RI, three RHI) by a factor w relative to the rest, in *both* directions, $w \in \{0.25, 0.5, 1, 2, 4, 8\}$. The equal split is the measured minimum of the resulting curve: at ε=0.8 the median two-hop RE runs $0.93 \to 0.84 \to 0.79 \to 1.01 \to 1.19 \to 1.79$ and at ε=3.2 $0.10 \to 0.14 \to 0.04 \to 0.14 \to 0.21 \to 0.42$ as w increases through that order, so both directions lose. Concentrating budget on the cross-hop objects starves the single/adjacent-hop marginals, and join-count utility is set by the noisiest factor in the join; starving the cross-hop object instead inflates its own error and erodes Δ-recovery (at ε=0.8, 0.44 at w=0.25 and 0.68 at w=0.5, against 0.87 at w=1, while for $w \geq 1$ recovery stays within the run-to-run spread, 0.80–1.05). Equal splitting is thus not a convenience but the empirical optimum of the sweep; workload-driven minimax allocation is left as future work. The technical appendix reports this ablation in full, as a white-box check that the default we adopt is a property of the join rather than of our tuning.

Other ablations. The route-A vs route-B comparison agreed as the design ablation is already realized by our two synthesis modes: Chain-NPM-direct (explicit cross-hop marginals, no latent mixture ≈ route A) vs Chain-NPM-mixture (latent mixture, route B). The mixture strictly improves Household⋈Individual RE (0.030 vs 0.084) and Δ-recovery (0.94 vs 0.87) at ε=3.2: the free EM step is what converts joint statistics into faithful within-group structure.

Mechanism ablations. (i) Removing the cross-hop marginals. Disabling all $R_0 \times R_2$ marginals (and the restricted triples) collapses Δ-recovery from 0.72–1.03 to −0.01–0.13 at every budget in both synthesis modes, while 2-hop RE degrades to the per-FK level (0.25–0.42): the cross-hop marginals are the sole carrier of the multi-hop signal, and the direct variant's residual (≤ 0.13) is exactly the observed mediation URBAN→OWN→EMP that per-FK models also enjoy. *(ii) EM iterations under a fixed budget.* Sweeping the EM iteration cap $T \in \{1, 2, 5, 10, 20\}$ at fixed (ε, δ) leaves Chain-NPM-mixture bit-identical—the mixture reaches its fixed point in one iteration on this benchmark—whereas the PrivLava-style accounting (fresh budget per iteration, per-iteration noise inflated by $\sqrt{T}$) degrades Household⋈Individual RE from 0.146 to 0.313 at ε=0.8 as T grows $1 \to 20$: budget dilution, not optimization, governs iterative EM. *(iii) Types and truncation.* Δ-recovery and within-group RE are nearly invariant to $k \in \{2, 3, 4, 6\}$ and $\tau \in \{4, 6, 8, 12\}$ (at τ=4 the 1-hop RE moves $0.030 \to 0.031$ and the 2-hop RE by about 5%); on this planted structure the latent types are symmetric and hence unidentifiable, so recovery is carried entirely by the cross-hop marginals—consistent with (i).

9.1 Efficiency

Chain-NPM's single budget pass of Q low-order marginals costs $O(Q \cdot \sum(n \cdot d))$ time and negligible memory, and the post-processing stage vectorizes over rows. On TPC-H SF=1, the full pipeline (42 marginals over 536K orders and 2.1M lineitems) completes in ≈ 2.2 minutes on one CPU; the deep three-table variant's synthesis takes ≈ 22–24 seconds. PrivPetal's official implementation needs 1.8–2.1 hours per budget on two A800 GPUs and 1.7–2.3 hours on one A800 for the deep variant (≈ 50× wall-clock gap on the standard rail, ≈ 300× on the deep one), and PrivLava's iterative EM carries the same budget cost. The gap follows directly from EM being post-processing rather than a budget-consuming loop.

Cost–effectiveness. Read as a cost–effect frontier, the same numbers place Chain-NPM alone in the region that preserves the effect (the technical appendix plots the frontier). On the four databases its retention error leaves the no-cross-table level at both budgets (0.011 at ε=0.8 and 0.009 at ε=3.2, against ≈ 1.0 for all five others) while it finishes in under a minute per budget, and PrivPetal is dominated at both: 75–119× the wall clock for the same effect level (1.0006, i.e. no effect retained). On the controlled benchmark, where all six methods run on one machine, Chain-NPM is also the cheapest point in the panel (2.0–2.2 s of driver wall clock per run, load included, against the per-table baseline's ≈ 2.5 s) and the only series above Δ-recovery 0.72. The mechanism thus does not buy effect preservation with wall clock; the axis on which it pays is query error at low budget (Section 6.3).

10 BUDGET-ALLOCATION ABLATION

The submission adopts the equal γ^2-split of the cross-hop marginals as its default and reports the sweep in prose; the figure below is the full ablation of that choice.

11 ANCESTOR-SCALE SWEEP

The submission quotes the endpoints of this sweep in the conclusion of Section 6.3 of the submission; the table below reports it in full.

12 COST–EFFECT FRONTIER

The submission states the cost–effectiveness conclusion in prose and refers here for the frontier itself; the efficiency discussion behind it is Section 9.

13 WALL-CLOCK COST PER BASIS

The row-level quota calibration used for every four-database Chain-NPM number in the paper is the released flag set –absorb 'pubn,rowleaf,szthresh,exactmarg,repair,growcap,szquota'; the default flags reproduce the plain release exactly (verified field-by-field).

The submission's efficiency section states the cost–accuracy conclusions and plots the frontier; the per-basis table below is the timing reference behind its claims. Timings compare within a row,

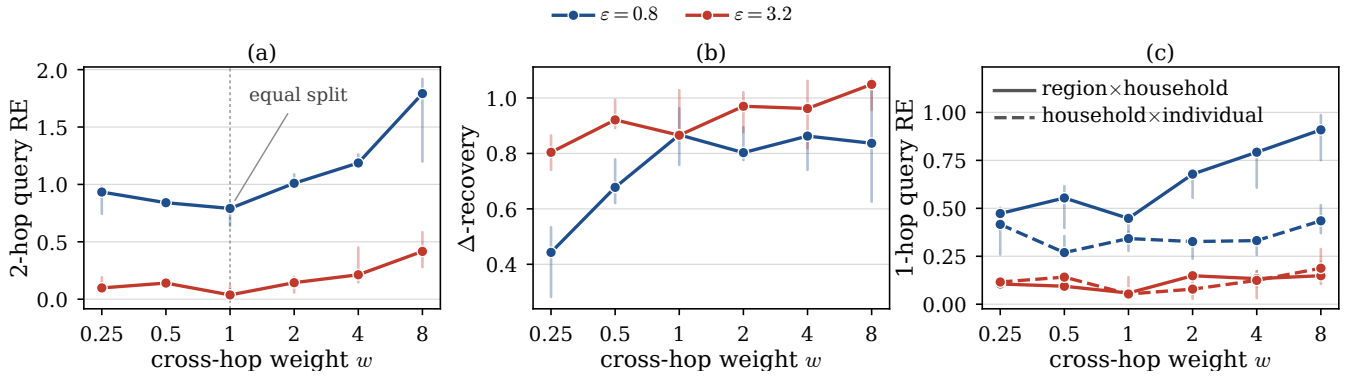


Figure 2: Budget-allocation ablation (CHAIN-NPM-direct; three runs per point, segments span the range; two budgets only, $\varepsilon \in \{0.8, 3.2\}$, so the curves carry no six-budget claim). The cross-hop weight w runs over $\{0.25, 0.5, 1, 2, 4, 8\}$ on a logarithmic axis. *(a)* Median two-hop query RE, lowest at w=1, the equal split, at both budgets. *(b)* Δ-recovery: unchanged within the run-to-run spread for $w \geq 1$ (0.80–1.05) and lower below it. *(c)* The two adjacent-hop classes (solid region×household, dashed household×individual), both worse at the two ends of the sweep than at w=1 and not monotone in between.

Table 7: Ancestor-scale sweep of the controlled benchmark (median over 3 runs; the same query workload and one stack per column): median two-hop query relative error by ancestor-table size, for the two CHAIN-NPM variants and the per-table lower bound (PRIVMRF). The smallest scale matches the deep chain's ancestor cardinality; there the per-individual marginals saturate (one-hop error 1.0) and no method recovers the planted effect, so that column is reported for completeness rather than compared.

	77	200	700	2500
ε=0.8				
PRIVMRF (per-table)	0.102	0.229	0.241	0.240
CHAIN-NPM-direct	1.644	0.791	0.444	0.293
CHAIN-NPM-mixture	1.525	0.772	0.438	0.319
ε=3.2				
PRIVMRF (per-table)	0.186	0.209	0.252	0.272
CHAIN-NPM-direct	0.426	0.037	0.032	0.015
CHAIN-NPM-mixture	0.447	0.114	0.008	0.018
ε=8.0				
PRIVMRF (per-table)	0.212	0.235	0.269	0.294
CHAIN-NPM-direct	0.150	0.062	0.043	0.030
CHAIN-NPM-mixture	0.219	0.024	0.039	0.030

never across rows, because each row is timed on its own reference chain implementation.

14 PLANTED-EFFECT RESULTS ON THE FOUR REAL DATABASES

Full planted-effect table. Table 9 is the full version of Figure 5 of the submission: retention error per dataset, protocol, budget and method, with the single-strength note and the (‡) marker explained in its caption.

15 NATURAL-MODE TABLES AND THE QUERY-CLASS STRATIFICATION

Natural-mode query error. Table 10 tabulates the natural-mode 2-hop query error of the same four databases, and Figure 4 gives the same view stratified by query class, for all six methods and all six budgets. The CHAIN-NPM rows are produced by the row-level quota-calibration arm of Section 6.1 of the submission: free post-processing that spends no budget and changes no released marginal. With it CHAIN-NPM is the most accurate method on the 2-hop class on 22 of the 24 dataset–budget cells (financial ties PRIVMRF at ε=0.4 and 0.8, the two gaps ≤ 0.02)—the deep chain's error collapses from 4.34 at ε=0.1 under the released arm to 0.28, and the V-shaped schemas lead their per-table baselines throughout (Table 10); on the class-stratified medians of Figure 4 it leads every cross-table class (2-hop, both 1-hop spans, and the two-parent class) at every budget, and the zero-hop class it was weakest in no longer shows the released arm's order-of-magnitude gap on the V-shaped schemas. Natural-mode query accuracy is therefore not a liability we qualify away: the correlation the per-FK family cannot represent is preserved (Table 9), and carrying it no longer costs

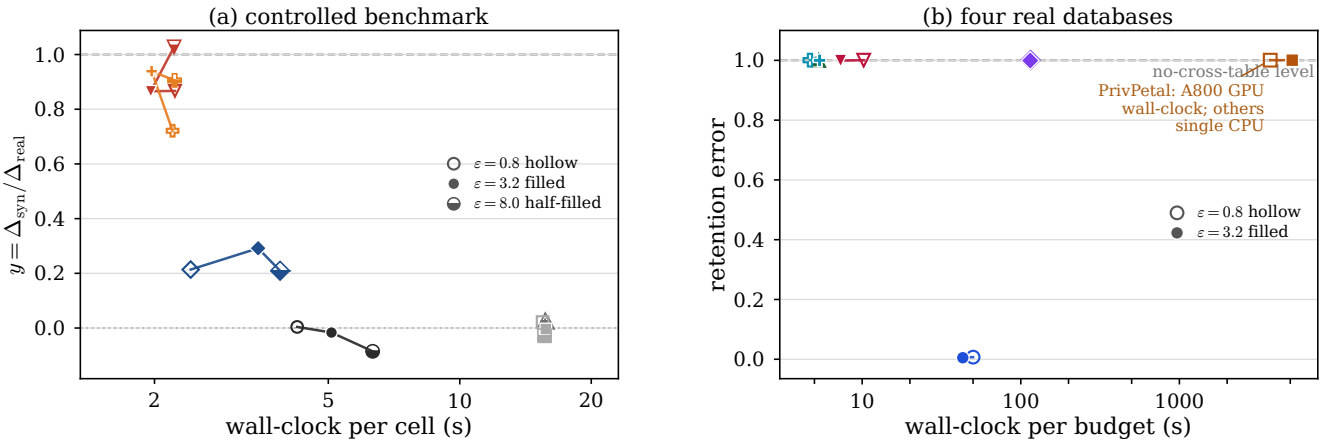


Figure 3: Cost against effect preservation. ***(a)*** **The controlled benchmark, the same 18 cells as Figure 6 of the submission:** y **is** $\Delta_{syn}/\Delta_{real}$ **(median over the three runs; dotted rule** y**=0, dashed rule** y**=1) and** x **the per-cell wall clock the driver recorded, so all cells come from one machine and one runner.** ***(b)*** **The four real databases:** x **is the median over the databases of the per-database median of the synthesis-phase wall clock (the evaluation is excluded), and** y **the retention error on the common basis—*plant_product* at** t**=0.316 and *plant_xor* at the XOR target value 0.8, ten seeds each, pooled per method, dataset and budget—with the dashed rule at the no-cross-table level. PrivPetal's** x **is that method's own wall clock on A800 GPUs while the other five are single-CPU numbers, the asymmetry of 8, which is conservative for Chain-NPM. Each cell's retention error is relative to that cell's own measured real effect, as everywhere in this paper; the six methods do not share one realized planted database (the four per-table and per-FK methods redraw it per seed), and pooling every cell onto a single denominator moves each mark by at most 0.018 and leaves Chain-NPM the lowest at both budgets. Marker fill encodes the budget (hollow** ε**=0.8, filled** ε**=3.2, and, in panel (a) only, half-filled** ε**=8.0), each panel carries its own legend, and panel (b) uses the palette of Figure 5 of the submission and 4.**

Table 8: Wall-clock cost per budget, stated per basis rather than netted out—the timing reference behind the submission's efficiency section, with the chain structure of Table 1 of the submission; each row is timed on its own reference chain implementation, so timings compare within a row, not across rows. The Chain-NPM column times the end-to-end pipeline on one CPU core (the TPC-H and ACS rows include restore and CSV output; the four-database rows isolate synthesis; evaluation is excluded); the PrivPetal column is that method's own wall-clock on A800 GPU(s). "—": not run on this benchmark.

Dataset	Private chain	Scale (rows)	ε grid	Chain-NPM	PrivPetal
Established benchmarks:					
Controlled (planted)	R←H←I	200 / ∼800 / ∼2400	0.8, 3.2, 8.0	≈1.1 s	—
TPC-H SF=1	Orders, Lineitem	536K / 2.1M	0.2, 3.2	134–136 s	1.8–2.1 h (2×A800)
TPC-H deep	Customer, Orders, Lineitem	150K / 536K / 2.1M	0.2, 3.2	≈22–24 s	1.7–2.3 h (1×A800)
ACS PUMS CA'22	Household, Person	151K / 383K	0.8, 3.2	≈20 s	hours (A800)
Four real databases:					
Financial (PKDD'99)	District←Account←Trans	77 / 4.5K / 1.06M	0.1–3.2	6–9 s	20–36 min
IMDb	Names←Principals→Titles	640K / 1.22M / 110K	0.1–3.2	115–117 s	56–106 min
Instacart	Products←Order-Products→Orders	38K / 1.52M / 150K	0.1–3.2	141–143 s	1.7–3.7 h
MovieLens-1M	Users←Ratings→Movies	6.0K / 1.00M / 3.9K	0.1–3.2	46–47 s	14–67 min

query accuracy, as the submission's planted-effect figure (Figure 5 of the submission) and Table 10 show.

16 POSITIONING OF REPRESENTATIVE MECHANISMS

The paper's related-work discussion carries the positioning; the table below is its tabular form.

Table 9: Planted-effect retention error (median over 10 seeds; 0 = effect survives intact, 1 = lost to the no-cross-table level, > 1 = distorted) on four real databases under two planting protocols. The deep chain (financial) reports the contrast Δ; the V-shaped schemas report the cross-parent cell contrast. Each *plant_product* cell is the range over the four planting strengths ($t = 0.1, 0.2, 0.316, 0.4$), so that no single strength is privileged, except for PrivPetal, whose planting protocol varies the privacy budget only and therefore contributes the single strength t=0.316; each *plant_xor* cell is a single value at the XOR target value 0.8, the only strength used. ‡the synthetic effect coincides across planting strengths, the signature of a method whose measured objects do not depend on the planting (see 6.2 for the two benchmark-side caveats that also produce it).

Dataset	ε	Chain-NPM	PrivMRF	PrivBayes	PB-PGM	PrivLava	PrivPetal
plant_product:							
financial	0.8	0.415–0.886	1.000–1.002	0.996–0.999	1.002–1.006	1.000–1.001‡	0.998
financial	3.2	0.175–0.346	1.000–1.002	0.997–0.999	1.001–1.004	0.996–0.999‡	1.000
imdb	0.8	0.007–0.011	1.003‡	0.999‡	1.001‡	1.011‡	0.223
imdb	3.2	0.005–0.007	0.997‡	0.998‡	0.999‡	1.005‡	0.055
instacart	0.8	0.007–0.018	0.993–0.998‡	0.997–0.999	0.988–0.997‡	0.996‡	1.003
instacart	3.2	0.005–0.020	1.003–1.013‡	0.997–0.999	1.000–1.001‡	0.997‡	1.000
movielens	0.8	0.007–0.021	1.003–1.007‡	1.003–1.007‡	0.998–0.999‡	1.002‡	0.994
movielens	3.2	0.004–0.021	0.995–0.998‡	1.000–1.001‡	0.999–1.000‡	0.998‡	1.010
plant_xor:							
financial	0.8	1.614	0.998	0.992	1.012	1.002	1.049
financial	3.2	0.846	0.997	0.989	1.006	0.993	1.048
imdb	0.8	0.010	1.003	0.999	1.001	1.010	0.249
imdb	3.2	0.017	0.997	0.998	0.999	1.006	0.302
instacart	0.8	0.001	1.001	1.000	1.001	1.001	1.002
instacart	3.2	0.001	0.999	1.001	1.000	1.001	1.000
movielens	0.8	0.003	1.001	1.001	1.000	1.001	1.000
movielens	3.2	0.002	0.999	1.000	1.000	0.999	0.999

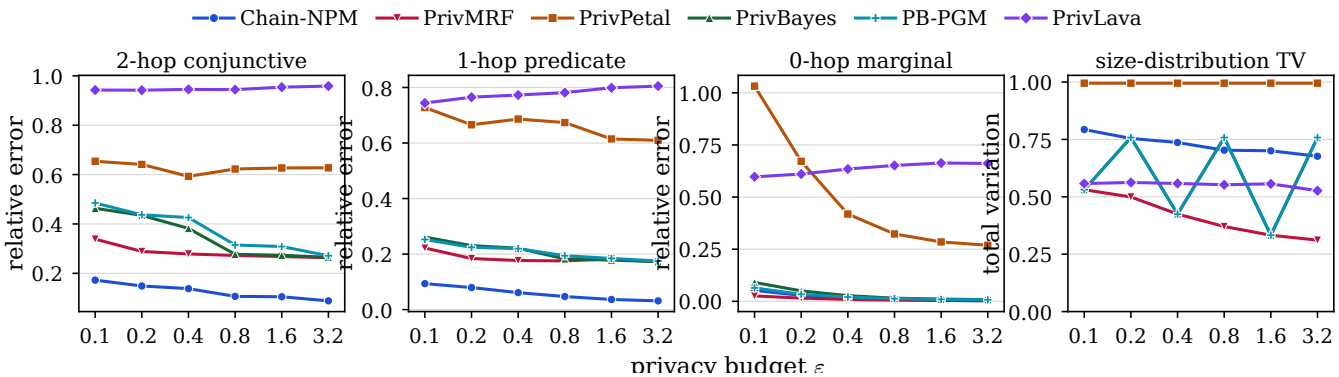


Figure 4: The natural-mode runs stratified by query class: 2-hop conjunctive, 1-hop predicate, 0-hop marginal, and size-distribution total variation. A class value averages that schema's member metrics arithmetically (the chain's two adjacent-hop objects, a V-shaped schema's two or three one-way marginals), then the 10 seeds are reduced to their median, then the four databases to theirs. Each curve is therefore the median over the datasets on which *all six* methods report that class, so no curve inside a panel is aggregated over a different basis than its neighbours; the size-distribution panel consequently reduces to the deep chain, the only database where all six report it. Vertical scales differ between panels and must not be compared across them.

17 THE THREE REAL-BENCHMARK RAILS

Figure 5 collects the three real-benchmark rails of Sections 7–8—standard TPC-H, the deep three-table variant, and ACS PUMS—in one place: each panel plots query relative error against the budgets that rail was actually run at, with one line per method and per hop class. The rails were run at two budgets each, so this summary is deliberately sparser than the four-database sweep of Table 10 and the class-stratified view below, and is meant to be read together with the submission's tables for the same rails, not instead of them.

REFERENCES

[1] Kaveh Alimohammadi, Hao Wang, Ojas Gulati, Akash Srivastava, and Navid Azizan. 2024. Differentially Private Synthetic Data Generation for Relational

Table 10: Natural mode (no planting): 2-hop large-query median relative error over 10 seeds, i.e. the cross-hop class (District×Trans for financial; the two-parent class for the V-shaped schemas). Lower is better. Chain-NPM rows use the row-level quota-calibration arm of Section 6.1 of the submission (the released default is bit-identical to the plain release); the arm spends no budget and changes no marginal, and it removes the released version's natural-mode query-accuracy gap on all four databases. The table is reported in full because the ranking is now part of the result, not a caveat to it.

Method	Dataset	ε=0.1	ε=0.2	ε=0.4	ε=0.8	ε=1.6	ε=3.2
Chain-NPM	financial	0.2774	0.2123	0.1868	0.0977	0.0969	0.0665
PrivMRF	financial	0.2875	0.2183	0.1698	0.0969	0.1357	0.1555
PrivBayes	financial	0.5180	0.4960	0.4017	0.1383	0.1734	0.1186
PB-PGM	financial	1.0000	0.8510	0.6907	0.2802	0.2667	0.1367
PrivLava	financial	0.3402	0.3798	0.3429	0.3686	0.3304	0.3430
PrivPetal	financial	0.2921	0.5083	0.7317	0.8421	0.9166	0.9430
Chain-NPM	imdb	0.1323	0.1286	0.1172	0.1157	0.1133	0.1106
PrivMRF	imdb	0.2050	0.2029	0.2049	0.2004	0.1954	0.1948
PrivBayes	imdb	0.2205	0.2221	0.2114	0.2028	0.1983	0.1974
PB-PGM	imdb	0.2230	0.2132	0.2101	0.2157	0.2024	0.2072
PrivLava	imdb	0.9992	0.9992	0.9993	0.9992	0.9991	0.9993
PrivPetal	imdb	0.6410	0.3064	0.3127	0.2925	0.3318	0.3642
Chain-NPM	instacart	0.0317	0.0275	0.0281	0.0254	0.0260	0.0254
PrivMRF	instacart	0.4912	0.4984	0.4780	0.5037	0.4915	0.4791
PrivBayes	instacart	0.6624	0.5591	0.5011	0.4919	0.4662	0.4709
PB-PGM	instacart	0.5016	0.4854	0.4692	0.4766	0.4821	0.4768
PrivLava	instacart	0.9044	0.9008	0.9083	0.9198	0.9337	0.9534
PrivPetal	instacart	1.2073	1.0324	0.4536	0.4443	0.4428	0.4383
Chain-NPM	movielens	0.2127	0.1690	0.1590	0.1345	0.1291	0.1301
PrivMRF	movielens	0.3894	0.3589	0.3521	0.3444	0.3409	0.3314
PrivBayes	movielens	0.4087	0.3736	0.3620	0.3515	0.3480	0.3322
PB-PGM	movielens	0.4679	0.3886	0.3829	0.3493	0.3503	0.3353
PrivLava	movielens	0.9805	0.9827	0.9819	0.9686	0.9742	0.9634
PrivPetal	movielens	0.6667	0.7739	0.8181	0.8007	0.8106	0.8166

Table 11: Positioning of representative DP relational synthesis mechanisms. *Cross-table object*: the statistical object each mechanism explicitly models across foreign keys; *budget pattern*: iterative budget consumption vs. a single measurement pass followed by free post-processing; *evaluated private depth*: the deepest private join path measured in the original evaluation; "—": not reported in the cited work.

Method	Cross-table object	Budget pattern	Evaluated private depth	Runtime order
PrivLava (SIGMOD'23)	per-FK latent models (EM); cross-hop only via estimated $\hat{Z}$ ($\leq$ 16 values)	iterative: each EM iteration consumes fresh budget	1 private FK (upstream tables public)	infeasible at EP scale
PrivPetal (SIGMOD'25)	per-FK permutation marginals; stages condition on synthetic parents; no joint object across two FKs	per-stage split (e.g., 0.5/0.25/0.25)	1 FK hop (query classes)	1.7–2.3 h per budget (A800)
Adapted single-table [1]	pairwise cross-table marginals only	—	—	—
Chain-NPM (ours)	joint chain marginals incl. cross-hop $R_0 \times R_2$; latent mixture refines only	single pass; EM as free post-processing	3 private tables	≈22–24 s (1 CPU)

Databases. arXiv:2405.18670.

[2] Kuntai Cai, Xiaokui Xiao, and Graham Cormode. 2023. PrivLava: Synthesizing Relational Data with Foreign Keys under Differential Privacy. In *Proc. ACM SIGMOD Int. Conf. Management of Data.* https://doi.org/10.1145/3589287

[3] Kuntai Cai, Xiaokui Xiao, and Yin Yang. 2025. PrivPetal: Relational Data Synthesis via Permutation Relations. In *Proc. ACM SIGMOD Int. Conf. Management of Data.*

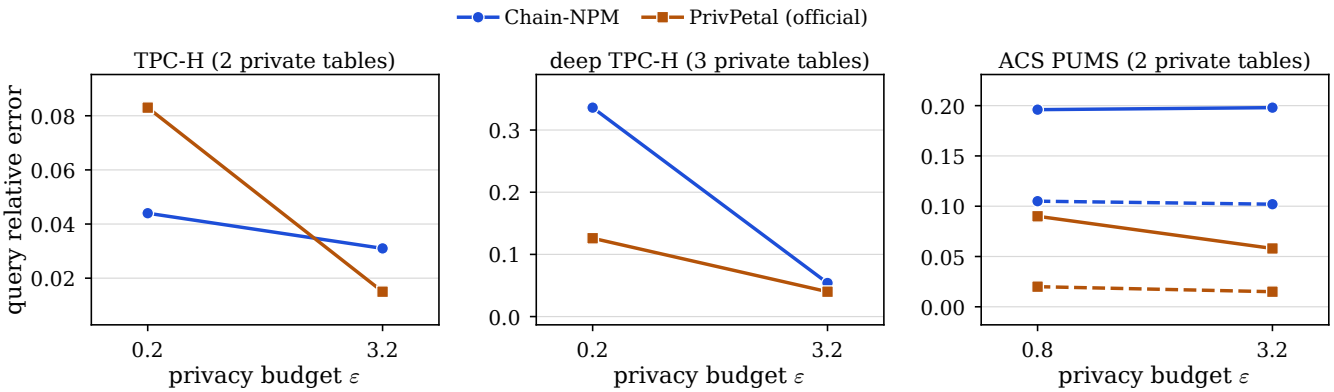


Figure 5: The three real-benchmark rails on one page: standard TPC-H (two private tables), the deep three-table variant, and ACS PUMS. Each panel uses that rail's own budgets ($\varepsilon \in \{0.2, 3.2\}$ for the two TPC-H variants, $\{0.8, 3.2\}$ for ACS) and plots query relative error; solid lines are the deepest hop that rail reports (two-hop for TPC-H and ACS, three-hop for the deep variant), and the dashed line is ACS's one-hop class. Values are the median relative errors tabulated in 4 and Table 6.